

Inferior STEMI with Bradycardia



Section I: Scenario Demographics

Scenario Title:	Inferior STEMI with 3 rd Degree Heart Block
Date of Development:	29/AUG/2016
Target Learning Group:	☐ Juniors (PGY 1 – 2) ☐ Seniors (PGY ≥ 3) ☒ All Groups

Section II: Scenario Developers

Scenario Developer(s):	Rob Woods
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Section III: Curriculum Integration

Learning Goals & Objectives	
Educational Goal:	To review the management of acute coronary syndrome, symptomatic bradycardia and cardiac arrest.
CRM Objectives:	1) Effectively lead a cardiac resuscitation. 2) Implement effective communication strategies during team management: a. Task Delegation & Closed Loop Communication b. Thinking out loud & summarizing
Medical Objectives:	1) Recognize inferior STEMI and heart block on ECG. 2) Initiate appropriate treatment for Acute Coronary Syndrome 3) Initiate appropriate treatment for symptomatic bradycardia in the setting of STEMI, including cardiac catheterization lab activation, transcutaneous pacing, and vasopressors 4) Recognize cardiac arrest and provide high quality ACLS care including: a. High quality CPR (30:2 compression rate) b. Use of ETCO₂ to guide resuscitation c. Minimizing pulse checks d. Appropriate 30:2 ratio of compressions to breaths prior to intubation

Case Summary: Brief Summary of Case Progression and Major Events

A 65-year-old female is brought to the ED with chest tightness and SOB. On arrival, she will be found to have an inferior STEMI with resultant 3rd degree heart block and hypotension. The team will be expected to initiate vasopressor support and transcutaneous pacing. However, prior to doing so, the patient will develop a VT arrest requiring ACLS care. After ROSC, the team will need to initiate transcutaneous pacing and activate the cath lab for definitive management.

References

Link MS, Berkow LC, Kudenchuk PJ, Halperin HR, Hess EP, Moitra VK, Neumar RW, O'Neil BJ, Paxton JH, Silvers SM, White RD. Part 7: adult advanced cardiovascular life support. *Circulation*. 2015 Nov 3;132(18 suppl 2):S444-64.

O'Connor RE, Al Ali AS, Brady WJ, Ghaemmaghami CA, Menon V, Welsford M, Shuster M. Part 9: acute coronary syndromes. *Circulation*. 2015 Nov 3;132(18 suppl 2):S483-500.



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Section IV: Scenario Script

A. Scenario Cast & Realism			
Patient:	☒ Computerized Mannequin	Realism:	☒ Conceptual
	☐ Mannequin		☐ Physical
	☐ Standardized Patient		☐ Emotional/Experiential
	☐ Hybrid		☐ Other:
	☐ Task Trainer		☐ N/A
Select most important dimension(s)			
Confederates	Brief Description of Role		
Bedside Nurse	Performs nursing tasks as delegated. Provides additional vitals if requested (blood sugar 6.4, Temp 37.1 °C). Frequently describes the patient as looking pale/gray and provides ongoing cues to team regarding abnormal vitals if team is not acting on them appropriately.		
B. Required Monitors			
☒ EKG Leads/Wires	☐ Temperature Probe	☐ Central Venous Line	
☒ NIBP Cuff	☒ Defibrillator Pads	☒ Capnography	
☒ Pulse Oximeter	☐ Arterial Line	☐ Other:	
C. Required Equipment			
☒ Gloves	☒ Nasal Prongs	☐ Scalpel	
☒ Stethoscope	☐ Venturi Mask	☐ Tube Thoracostomy Kit	
☒ Defibrillator	☒ Non-Rebreather Mask	☐ Cricothyroidotomy Kit	
☒ IV Bags/Lines	☒ Bag Valve Mask	☐ Thoracotomy Kit	
☒ IV Push Medications	☒ Laryngoscope	☐ Central Line Kit	
☐ PO Tabs	☒ Video Assisted Laryngoscope	☐ Arterial Line Kit	
☐ Blood Products	☒ ET Tubes	☒ Bedside Ultrasound	
☐ Intraosseous Set-up	☒ LMA	☐ Other: C-spine collar	
D. Moulage			
Mannequin to be described or dressed as 65-year-old female who is obese (height 5'3", weight ~ 90kg). Non-rebreather mask should be in place.			
E. Approximate Timing			
Set-Up:	20 min	Scenario:	15-20 min
		Debriefing:	20-30 min

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Section V: Patient Data and Baseline State

A. Clinical Vignette: To Read Aloud at Beginning of Case			
To be stated by the bedside nurse: "This 65-year-old woman came in with 1 hour of chest pressure and SOB. Her O2 sats were 84% on RA at triage, and they are now 90% with a non-rebreather mask. She's also bradycardic at 30 and hypotensive at 77/40."			
B. Patient Profile and History			
Patient Name: Hannah Van Vuuren		Age: 65	Weight: 90kg (5'3" or 160cm)
Gender: ☐ M ☒ F		Code Status: Presumed full.	
Chief Complaint: Chest tightness and shortness of breath			
History of Presenting Illness: Husband brought her to the hospital when she started complaining of SOB and chest tightness. Well prior to 1 hour ago. She feels like someone is squeezing her chest.			
Past Medical History:	HTN, Smoker, Dyslipidemia, Sleep Apnea	Home Medications: Cilazapril, Atorvastatin No home CPAP (could not tolerate the machine)	
Allergies: None			
Social History: Lives with her husband, 1 PPD smoker for the last 45 years			
Family History: None			
Review of Systems:	CNS:	NA	
	HEENT:	NA	
	CVS:	NA	
	RESP:	Admitted with pneumonia last winter, frequent chest infections	
	GI:	NA	
	GU:	NA	
	MSK:	NA	INT:
C. Baseline Simulator State and Physical Exam			
☐ No Monitor Display		☐ Monitor On, no data displayed	☒ Monitor on Standard Display
HR: 32/min	BP: 77/40	RR: 32/min	O ₂ SAT: 90% on NRB
Rhythm: 3 rd Deg AVB	T: 37.1°C	Glucose: 6.4 mmol/L	GCS: 14 (E4 V4 M6)
General Status: Sitting upright, looks distressed, having trouble answering questions, diaphoretic.			
CNS:	PERLA, 3mm bilaterally.		
HEENT:	Nil.		
CVS:	Irregular heart sounds, no murmur. Elevated JVP.		
RESP:	Tachypneic, coarse crackles throughout, accessory muscle use.		
ABDO:	Nil.		
GU:	Nil.		
MSK:	Nil.	SKIN:	Nil.

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Section VI: Case Progression

Scenario States, Modifiers and Triggers			
Patient State	Patient Status	Learner Actions, Modifiers & Triggers to Move to Next State	
1. Baseline State Rhythm: 3 rd degree AVB HR: 32/min BP: 77/40 RR: 32/min O ₂ SAT: 90% NRB T: 37.1°C	GCS 14.	<u>Learner Actions</u> ☐ Primary Survey, coordinate team ☐ IV access, O ₂ , monitors ☐ STAT 12 lead ECG ☐ Apply defib pads ☐ Brief history and physical exam ☐ IV NS 500ml bolus ☐ ± Portable CXR ☐ Call cath lab	<u>Modifiers</u> <u>Triggers</u> - 3 min into state → 2. More Drowsy
2. More Drowsy HR: 30/min (long 6-10 second pauses) BP: 66/36 RR: 32/min O ₂ SAT: 90% NRB	GCS, now 12/15 (E3, V3, M6). Nurse to start state by saying "she seems more drowsy."	<u>Learner Actions</u> ☐ Atropine 0.5mg iv push ☐ Initiate trans-cutaneous pacing ☐ ± Start epinephrine infusion (2-10mcg/min) ☐ Consider BiPAP as bridge to intubation	<u>Modifiers</u> - Atropine given → no effect <u>Triggers</u> - About to initiate pacing OR 4 min into state → 3. VT Arrest
3. Cardiac Arrest Rhythm: VT HR: 200 BP: -/- O ₂ SAT: 0% ETCO ₂ : 10	Pulseless unconscious and pulseless	<u>Learner Actions</u> ☐ Immediate high quality CPR ☐ CPR ratio (30:2 until intubation) ☐ Shock VT at 200J ☐ Minimize peri-shock pause ☐ Give epinephrine 1mg IV q 3-5 minutes ☐ Amiodarone 300mg iv ☐ ± Intubation	<u>Modifiers</u> - Poor quality CPR → ETCO ₂ 10 and nurse to comment - High quality CPR → ETCO ₂ 20 <u>Triggers</u> - 1 shock/2 rounds CPR → 4. ROSC
4. ROSC Rhythm: 3 rd degree AVB HR: 30 BP: 72/44 O ₂ SAT: 86% NRB	GCS 3.	<u>Learner Actions</u> ☐ Repeat ECG ☐ Atropine 0.5mg iv push ☐ Start epinephrine infusion (2-10mcg/min) ☐ Initiate trans-cutaneous pacing ☐ Consult Cardiology for PCI	<u>Modifiers</u> - Atropine given → no effect <u>Triggers</u> - Pacing initiated or epi infusion started → 5. Resolution
5. Resolution HR 80 BP 100/60 O ₂ Sat → 94%	Unchanged.	<u>Learner Actions</u> ☐ Maintain epi infusion ☐ Call cath lab (if not yet done) ☐ Intubation to facilitate PCI ☐ OG insertion to administer ASA and ticagrelor ☐ Administer 4000 units iv heparin ☐ Prepare for transvenous pacemaker insertion	End Case PRN with "the cath lab is ready!"

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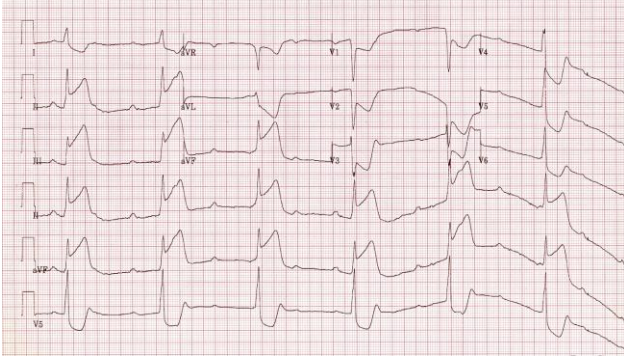
Section VII: Supporting Documents, Laboratory Results, & Multimedia

Laboratory Results

No blood work is provided during the case.

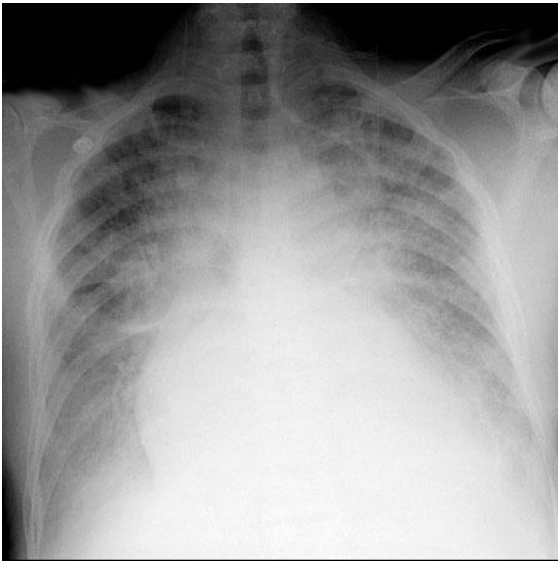
Images (ECGs, CXRs, etc.)

ECG – Inferior STEMI with 3rd degree AV Block



ECG source: <http://lifeinthefastlane.com/ecg-library/basics/inferior-stemi/>

CXR – CHF



CXR source: <https://www.meded.virginia.edu/courses/rad/cxr/pathology2Bchest.html>

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Section VIII: Debriefing Guide

General Debriefing Plan	
☐ Individual	☒ Group
☐ With Video	☒ Without Video
Objectives	
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Key Prompting Questions	
1) Did you feel that your team leader had control over the room? Did he or she ask for input? 2) What was the cause of this patient's abnormal vital signs and level of consciousness? 3) What interventions are going to improve this patient's status? 4) When should you consider using BiPAP in the setting of heart failure? 5) What can happen if you intubate and ventilate a patient whose blood pressure is quite low? 6) What options are available for treating bradycardia? What do the ACLS guidelines indicate should be the initial treatments in unstable bradycardia?	
Key Take Home Points	
Recognition of bradycardia as complication of STEMI (and need for urgent PCI)	
Management of unstable bradycardia with transcutaneous pacing and vasopressors	
High quality CPR and arrest management is key to survival	